

Exp No: 5 – Data Collection and Initial Exploration

Aim

To collect, load, and perform initial exploration of the diabetes dataset.

Description

The diabetes dataset is loaded using Pandas. The dataset is explored using `head()`, `info()` and `describe()`. Histograms are used to visualize the distribution of each feature and a pair plot is used to visualize relationships between the features.

Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

url = "https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv"

columns = ['Pregnancies','Glucose','BloodPressure','SkinThickness',
           'Insulin','BMI','DiabetesPedigreeFunction','Age','Outcome']

db = pd.read_csv(url, names=columns)

print(db.head())

print(db.info())
print(db.describe())

db.hist(bins=50, figsize=(20,15))
plt.show()

sns.pairplot(db)
plt.show()
Output:
```

```

Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin  BMI  \
0           6      148           72           35         0  33.6
1           1       85           66           29         0  26.6
2           8      183           64           0          0  23.3
3           1       89           66           23         94  28.1
4           0      137           40           35        168  43.1

```

```

DiabetesPedigreeFunction  Age  Outcome
0           0.627      50         1
1           0.351      31         0
2           0.672      32         1
3           0.167      21         0
4           2.288      33         1

```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 768 entries, 0 to 767
```

```
Data columns (total 9 columns):
```

```

#   Column                                Non-Null Count  Dtype
---  -
0   Pregnancies                          768 non-null    int64
1   Glucose                              768 non-null    int64
2   BloodPressure                        768 non-null    int64
3   SkinThickness                        768 non-null    int64
4   Insulin                              768 non-null    int64
5   BMI                                  768 non-null    float64
6   DiabetesPedigreeFunction             768 non-null    float64
7   Age                                  768 non-null    int64
8   Outcome                              768 non-null    int64

```

```
dtypes: float64(2), int64(7)
```

```
memory usage: 54.1 KB
```

```
None
```

```

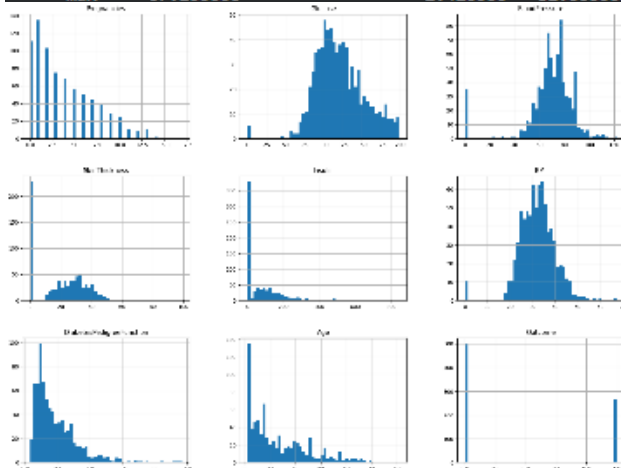
Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin  \
count  768.000000  768.000000  768.000000  768.000000  768.000000
mean     3.845052  120.894531  69.105469  20.536458  79.799479
std     3.369578   31.972618  19.355807  15.952218  115.244002
min     0.000000   0.000000   0.000000   0.000000   0.000000
25%     1.000000   99.000000   62.000000   0.000000   0.000000
50%     3.000000  117.000000   72.000000   23.000000  30.500000
75%     6.000000  140.250000   80.000000   32.000000  127.250000
max    17.000000  199.000000  122.000000   99.000000  846.000000

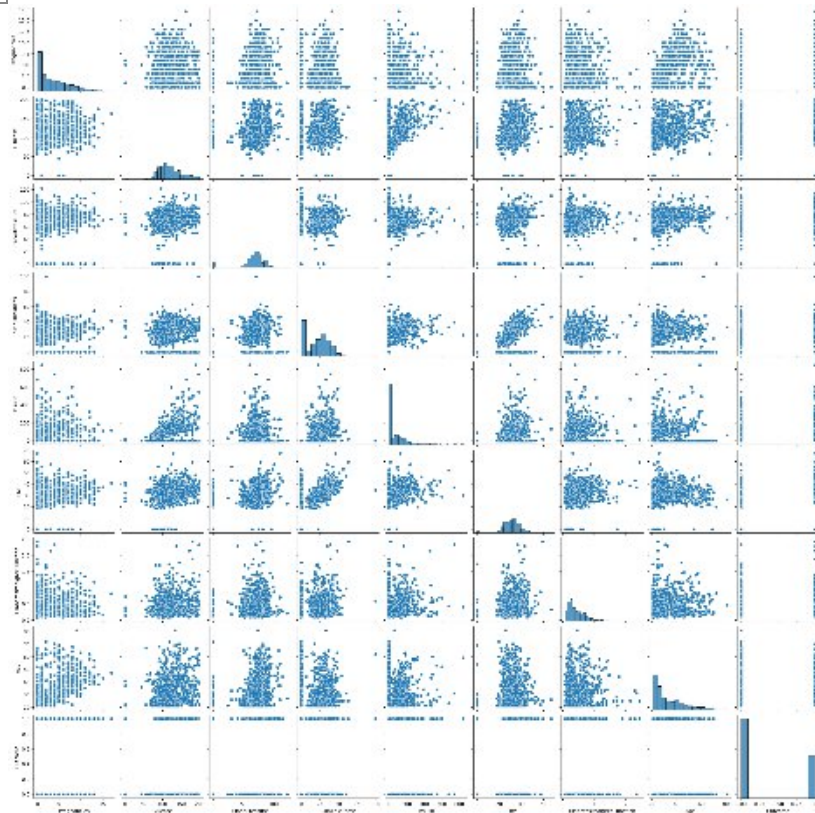
```

```

BMI  DiabetesPedigreeFunction  Age  Outcome
count  768.000000  768.000000  768.000000  768.000000
mean    31.992578    0.471876  33.240885    0.348958
std     7.884160    0.331329  11.760232    0.476951
min     0.000000    0.078000  21.000000    0.000000
25%    27.300000    0.243750  24.000000    0.000000
50%    32.000000    0.372500  29.000000    0.000000
75%    36.600000    0.626250  41.000000    1.000000
max    67.100000    2.420000  81.000000    1.000000

```





Result

The diabetes dataset was successfully loaded and initially explored. The first few records, dataset structure and summary statistics were displayed. Histograms showed the distribution of each feature, and the pair plot showed the relationships between the variables.